

Xiaoyu Ma

xiuyunMarx@gmail.com • xiaoyu_ma@sjtu.edu.cn • github.com/xiuyunMarx

Research Interests

AI Infrastructure, LLM Serving & Agents, GPU Programming, RDMA Networking, Cloud Computing, Programming Language.

Education

Shanghai Jiao Tong University, China

B.Eng. in Computer Science and Engineering

2023 – 2027 (expected)

Core GPA: 3.79/4.0 (90.38/100)

- **Selected Coursework:**

- *Systems & Architecture*: Modern Operating Systems (99), Computer Architecture Experiments (100), Computer Architecture (95), Compilers (92), Computer Organization (92)
- *AI, Theory & Networks*: Natural Language Processing (98), Network Security Technology (98), Machine Learning (94), Basics of Cryptography (94), Theory of Computation (93), Algorithms and Complexity (92)

- **Standardized Tests:** TOEFL iBT: 110 / 120

Research Experience

Jaseci Labs, University of Michigan

July 2026 – Present *Research Intern*

- Working on byLLM, a framework for integrating LLMs into programming language runtimes.

SLO-Aware Proactive History Compression for Long-History Generative Recommendation Serving

First Author (Submitted to ACM SoCC '26)

- Designed and implemented of an SLO-aware GPU multiplexing system featuring asynchronous history compression for long-history generative recommendation serving under fluctuating workloads.
- Designed a proactive compression scheduling policy that overlaps history compression with online inference to maintain goodput.

ZAP: Adaptive Templating for Fast Startup of LLM Functions on GPUs

- Contributed to the asynchronous model weight-loading pipeline with multi-threading optimizations, and resolved critical-path correctness issues in the serving runtime.
- Implemented heterogeneous GPU peer-to-peer (P2P) communication and RDMA-based remote model transfer; benchmarked startup latency and execution overhead.

Course Projects

- **Tiny File System:** Built an inode-based distributed file system over TCP with shell-like operations, write-ahead logging for crash consistency, and concurrent multi-user client handling.
- **RAG Course Assistant Agent:** Built a multimodal (text + image) RAG agent with vector databases, designing preprocessing, chunking, and vectorization pipelines to improve retrieval accuracy.

Skills

- **Programming Languages:** C/C++, CUDA, Python, Jac
- **Areas of Expertise:** AI Infrastructure, GPU Memory Management, LLM Serving Systems

Scholarships & Awards

SJTU Academic Excellence Scholarship (Class C)

2025